

Aditya Rajan

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Statistics & Computer Science, Minor in Mathematics GPA: 3.93/4.00 (Dean's List Fall '24) August 2023 – May 2026

EXPERIENCE

Machine Learning Research Intern

IDiA Lab

June 2025 – Present

University of Wisconsin-Madison

- Developed a novel **CLIP-based vision-language model** for stroke lesion segmentation, achieving a **Dice score of 0.81** and outperforming existing baselines like nnU-Net and Swin-UNETR
- Explored domain-agnostic MRI data augmentation approaches using **physics-informed generative models**.
- Researched, wrote, and compiled a scientific abstract evaluating the CLIP model's performance, submitted to **SPIE Medical Imaging 2025**.

Undergraduate Researcher

Alam Lab for Computational Genomics

January 2025 – Present

University of Illinois at Urbana-Champaign

- Conducted **S. pneumoniae & S. aureus genome data analysis** from High-Throughput Sequencing technologies.
- Designed **large-scale data pipelines** and **k-mer search algorithms** for prophage detection in **2000+ annotated genome samples**.
- Developed **novel transformer-based ML solutions** to examine antimicrobial resistance and virulence in pathogenic bacteria.

Machine Learning Intern

Madhya Pradesh State Electronics Development Corporation

May 2024 – June 2024

Bhopal, MP, India

- Analyzed family registration data to integrate registration processes with the state government's **Single Citizen Database** (Samagra ID) for **24 million+ registered citizens**.
- Implemented **video-based KYC** for the ongoing Samagra e-KYC project using **computer vision models in Python and C++**.
- Automated state and national ID linking for state citizens, **improving data processing efficiency by 15%**.

PROJECTS

CRISPR-Cas9 Off-Target Prediction | *Python – Scikit-learn, Pandas, NumPy*

Jan 2025 – Present

- Built a **pipeline** for **340,000+ gRNA sequences**, extracting features (**GC content, edit distance, sequence similarity**).
- Developed ML models (**Logistic Regression, XGBoost, Transformer**) with **98% accuracy** and **ROC-AUC > 0.96**.
- Implemented **sequence-to-numeric encodings** to preserve biological signal.

Network Port Scanning CLI Tool | *Rust – Clap, Tokio*

May 2024 – Jul 2024

- Developed a **multithreaded CLI scanner** (TCP, HTTP), achieving **30% faster scans** than traditional tools.
- Applied **async Rust** for performance; added **robust logging and error-handling**.
- Designed a **modular command-line interface** with configurable options for scan modes and verbosity.

Interactive A* Search Algorithm Visualizer | *Python – Pygame*

Aug 2023 – Sep 2023

- Created an **interactive tool** to visualize the **A* Pathfinding Algorithm**.
- Implemented **single source shortest-path search** in maze-like grids with **sub-second runtime on average**.
- Built a **responsive UI** with mouse event handling via Pygame.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, PostgreSQL, Rust, R

Frameworks/Tools: Git, Docker, VSCode, Android Studio, SLURM, Conda

Libraries: Tensorflow, scikit-learn, PyTorch, pandas, NumPy, Matplotlib, NiBabel, BioPython